

MICHELLE WAGNER

Rego Park, NY 11374 • (347) 456-5124 • mwagner000@citymail.cuny.edu

<https://www.linkedin.com/in/michelle-may-wagner>

Skills

- Satellite data processing and geospatial analysis (SNAP, QGIS/ArcGIS, Google Earth)
- Calibration and validation of ocean color remote sensing (Sea DASS, Hydrolyte)
- Atmospheric Reanalysis & Climate Modelling (EMARS, MGCM)
- Programming (Python, MATLAB, RStudio, C++, Arduino)

Work Experiences and Professional Appointments:

Adjunct Lab Instructor – 08/2025 to Current

Research Foundation City University of New York (RF CUNY) at City College (CCNY) -Harlem, NY

- Instructed and supervised undergraduate laboratory sessions, preparing and delivering course materials, guiding students in applying theoretical concepts to hands-on activities, and maintaining a supportive and organized lab environment.
- Assessed student performance through grading and feedback, and collaborated with faculty to ensure alignment between lab and lecture components.

NOAA EPP/MSI CESSRST-II Cohort 3 Graduate Fellow – 03/01/2025 to Current

National Oceanic and Atmospheric Administration (NOAA) Cooperative Center for Earth System Sciences and Remote Sensing Technologies (CESSRST)

- Contributed professional research to the NOAA Educational Partnership Program with Minority Serving Institutions (EPP/MSI) *Weather and Atmosphere* Research Thematic Area

Environmental Educator/Air Quality Research Analyst, 05/2024 to Current

Eastern Queens Alliance Inc., Idlewild Environmental Center – Springfield Gardens, NY

- Designed presentations and communicated complex environmental concepts to a diverse audience, engaging students, community members and local stakeholders
- Spearheaded EQA's first ever remote sensing workshop series with hands-on experiments and guided computer-based data analytic instruction
- Conducted detailed air quality monitoring and analysis and contributed to the writing of grant proposals, and publications

Research Intern, 05/2023 to 08/2023

Tzortziou Bio-optics Lab – Center for Discovery and Innovation, Harlem, NY

- Utilized ground based and satellite remote sensing bio-optical tools to assess the impact of natural and anthropogenic stressors on coastal systems
- Assisted with calibration and validation of ocean color remote sensing data and contributed to algorithm optimization associated with the development of level 3 data products, using statistical and visual programs such as Sea DASS, Hydrolyte and MATLAB
- Analyzed and evaluated ecological trends in urban estuaries and other threatened ecosystems (i.e. Long Island Sound, Chesapeake Bay, Norton Sound)
- Routinely participated in fieldwork, preparing samples for analysis, maintaining cleanliness of lab equipment and ensuring adherence to lab safety standards

Undergraduate Research Intern, 05/2022 to 08/2022

Research Experiences for Undergraduates at ASU – Alabama State University

- Conducted comprehensive experiments to enhance the accuracy of Laser-Induced Breakdown Spectroscopy (LIBS) methodologies through the application of gold nanoparticles
- Prepared detailed reports and visualizations summarizing research findings and recommendations for further actions and presented final findings to interdepartmental staff, community members and stakeholders at REU research symposium

Group Leader & STEM Coordinator, 02/2017 to 12/2017

The Beacon Program at M.S. 226 – South Ozone Pk, NY

- Prepared after-school enrichment activities, fostering classroom engagement and provided tutoring services in both one on one and group settings
- Developed and implemented engaging STEM lessons for a variety of grade levels

Student Assistant, Summer 2016

Center for Prevention and Outreach (CPO) – Stony Brook, NY

- Assisted the main office by coordinating appointments, maintaining organization, answering phones, greeting incoming office visitors and maintaining confidentiality of patient files.
- Supported administrative staff with clerical tasks such as data entry tasks, ensuring accuracy in record-keeping systems.

Education

Master of Science: Earth and Atmospheric Science, Expected in 8/2023

City College of NY - Harlem, NY

Bachelor of Science: Earth Systems Science, Ecosystems concentration 05/2023

City College of NY - Harlem, NY

Honors and Awards

- Awarded funding by the Ocean Carbon Biogeochemical Program to attend the Summer 2024 Satellite Remote Sensing Training Program, enhancing expertise in environmental science and supporting professional development
- Received the Jeffrey Steiner award for excellence in undergraduate research in earth and atmospheric sciences from the City College of NY (2023)

Professional Trainings:

- Responsible Conduct of Research (RCR) for City University of New York (CUNY) Researchers, Collaborative Institutional Training Initiative (CITI) Program
Completion Date: 13-Apr-2025 | Expires: 13-Apr-2029
- NASA's Applied Remote Sensing Training (ARSET) Program
 - Fundamentals of Remote Sensing
 - NASA's Observing Fleet
 - Calculating Spectral Indices for Land and Aquatic Applications Using QGIS

Volunteer Experience

- Phytoplankton Monitoring Network (PMN)
- STEMx Tutoring
- The Compost Collective
- NYC Parks Stewardship
- Discovery Room, AMNH
- Gowanus Canal Conservancy

Presentations and Reports:

Wagner, M., Barnett, C., & Vig, K. (2022, August 30). *Examining the influence of gold nanoparticles on signal enhancement in LIBS technology*. Research & Training in Multidisciplinary Field of Regenerative Sciences for Undergraduates Symposium, Alabama State University

Bryan, J., Wagner, M., Ahmed, E., Holder, J., Duncan, L., Akeju, B. A., Mendela, M., & Roberts-Semple, D. (2015, January 15). *Assessing airport-related noise and air pollution levels in Southeast Queens* Eastern Queens Alliance Inc., DEC Community Impact Grant Report

Dominique Jenssen,¹ Minsun Lee,² Charlotte Rhoads,² Joaquim Goes,¹ Maria Tzortziou,¹ Michelle Wagner,² Kyle Turner², (May 2023) ¹Columbia University, ²City College of New York, *A Space-based Retrieval Approach for Red-brown Tides in Long Island Sound*, Lamont- Doherty Earth Observatory